MATH 3042- Assignment 5- Solutions

MATH 3042- Assignment 5 Full Name:

When you use R to look up probability, please write the filled-in R command along with the result.

1. The time (in hours) that a patient waits in a hospital emergency room before being seen by a doctor is exponentially distributed, with a mean of 2.25 hours.

   a. [2] What is the probability that a patient is seen by a doctor within 1 hour of arrival?

Let x be the wait time a random patient. $X \sim \exp(2.25)$

$P(x < 1) = 1 - e^{-\frac{1}{2.25}} = 35.88\%$

   b. [2] What is the probability that a patient waits between 2 and 4 hours?

$P(2 < x < 4) = \left(1 - e^{-\frac{4}{2.25}}\right) - \left(1 - e^{-\frac{2}{2.25}}\right) = 24.21\%$

   c. [2] What is the 95th percentile of wait times?

$1 - e^{-\frac{t}{2.25}} = 0.95, \qquad 0.05 = e^{-\frac{t}{2.25}}, \qquad \ln 0.05 = -\frac{t}{2.25}, \; t = 6.74\ hours$

2. The elevator in a building takes between 0 and 22.5 seconds to arrive after the button is pushed, with all times in that interval being equally likely.

   a. [2] What is the probability that wait time will be less than 12 seconds for the elevator to arrive the next time the elevator button is pushed?

Let x be the wait time of random person for this elevator. $X \sim unif(0,22.5)$ this is a scarry elevator :)

$P(x < 12) = \frac{12}{22.5} = 53.3\%$

   b. [2] If I took the elevator two independent times, what is the probability that I had to wait more than 20 seconds **both** times?

$P(x > 20) = \frac{2.5}{22.5} = 11.11\%$

$P(k = 2) = C(2,2)(0.1111)^2(0.8889)^0 = 0.1111^2 = 0.012346 \approx 1.23\%$

   c. [2] What is an unusually long wait time? Use $z > 2$ criteria

Uniform mean: $\frac{0+22.5}{2} = 11.25$

Uniform $\sigma^2 = \frac{(22.5-0)^2}{12} = 42.19$

Uniform $\sigma = \sqrt{42.19} = 6.5$

$\pm 2 = \frac{x - 11.25}{6.5}$, $x_{low} = -1.74$ $x_{high} = 24.24$

Both are out of range of x. There are no unusual values.

3. The systolic blood pressure of healthy adults is normally distributed, with a mean of 112 mmHg and a standard deviation of 10 mmHg.

   a. [2] What is the probability that a randomly-chosen healthy adult has a systolic blood pressure between 110 mmHg and 130 mmHg?

Let x be the blood pressure (BP) of a random healthy adult. $X \sim norm(112, 10)$
$P(110 < x < 130) = pnorm(130, 112, 10) - pnorm(110, 112, 10) = 54.33\%$

   b. [2] What is the probability that a randomly-chosen healthy adult has a systolic blood pressure greater than 130 mmHg?

$P(x > 130) = 1 - pnorm(130, 112, 10) = 3.59\%$

   c. [2] what values mark the middle 80% of systolic blood pressure?

$qnorm(0.10) = -1.281552$

$\pm 1.281552 = \frac{x - 112}{10}, \quad x_{low} = 99.2\ mmHg \quad x_{high} = 124.8\ mmHg$

4. The response time for requests to a web server follows a normal distribution with a mean of 180 milliseconds and a standard deviation of 35 milliseconds.

   a. [2] If the server displays a timeout warning for any request that takes longer than 240 milliseconds, what percentage of requests will trigger this warning?

Let x be the response time for a random request to a web server. $X \sim norm(180, 35)$
$P(x > 240) = 1 - pnorm(240, 180, 35) = 4.32\%$

   b. [2] The development team wants to set a performance benchmark such that only the slowest 5% of requests exceed it. What response time threshold should they use for this benchmark?

Slowest 5%: the 5th percentile
$P_5: qnorm(0.05) = -1.64485$
$-1.64485 = \frac{x - 180}{35}, \qquad x = 122.43ms$